Worksheet 4

Instructions

- Include your full name and Coyote ID on the top right of the first page.
- Submit completed worksheets to Gradescope (no tagging required).
- Submit before 11:59 PM on Wednesday, September 30, 2026

Physics Problem Solving Tips

- Draw a diagram (if possible).
- Write down what you know.
- Write down what you want to find.
- First solve symbolically; plug in values at the end.
- Check to make sure your units are correct.
- If you plug in values, check order of magnitude.

Problems

This worksheet covers kinematics in 2D — circular motion..

1. Your roommate is working on his bicycle and has the bike up-side-down. He spends the $60\ \text{cm}$ diameter wheel, and you notice that a pebble stuck in the tread goes by three times every second. What are the pebble's speed and acceleration?

SOLVE: The pebble's angular velocity $\omega = (3.0\ \text{rev/s})(2\pi\ \text{rad/rev}) = 18.9\ \text{rad/s}$. The speed of the pebble as it moves around a circle of radius $r = 30\ \text{cm} = 0.30\ \text{m}$ is

$$v = \omega r = (18.9\ \text{rad/s})(0.30\ \text{m}) = 5.7\ \text{m/s}$$

The radial acceleration is

$$a_r = \frac{v^2}{r} = \frac{(5.7\ \text{m/s})^2}{0.30\ \text{m}} = 108\ \text{m/s}^2$$

2. A bicycle wheel is rotating at $50\ \text{rpm}$ when the cyclist begins to pedal harder, giving the wheel a constant angular acceleration of $0.50\ \text{rads/s}^2$.
   a. What is the wheel's angular velocity, in rpm, $10\ \text{s}$ later?
   b. How many revolutions does the wheel make during this time?

MODEL: The wheel is in nonuniform circular motion.

VISUALIZE:

Known
$\omega_0 = 50$ rpm
$\alpha = 0.50$ rad/s^2
$\theta_i = 0$

Find
ω_f θ_f

Pictorial representation



SOLVE: **(a)** Express ω_i in rad/s:

$$\omega_i = (50\ \text{rev/min})\left(\frac{\text{min}}{60\ \text{s}}\right)\left(\frac{2\pi\ \text{rad}}{\text{rev}}\right) \Rightarrow 5.2\ \text{rad/s}$$

After 10 s, $\omega_f = \omega_i + \alpha\Delta t \Rightarrow \omega_f = 5.2\ \text{rad/s} + (0.50\ \text{rad/s}^2)(10\ \text{s})^2 = 55\ \text{rad/s}$. Converting to rpm,

$$(55\ \text{rad/s})\left(\frac{60\ \text{s}}{\text{min}}\right)\left(\frac{\text{rev}}{2\pi\ \text{rad}}\right) = 53\ \text{rpm}$$

(b) In 10 s, the wheel has turned a number of radians

$$\theta_f = \theta_i + \omega_0\Delta t + \tfrac{1}{2}\alpha\Delta t^2 \Rightarrow \theta_f = 0\ \text{rad/sec} + (5.2\ \text{rad/s})(10\ \text{s}) + \tfrac{1}{2}(0.50\ \text{rad/s}^2)(10\ \text{s})^2 = 77\ \text{radians}.$$

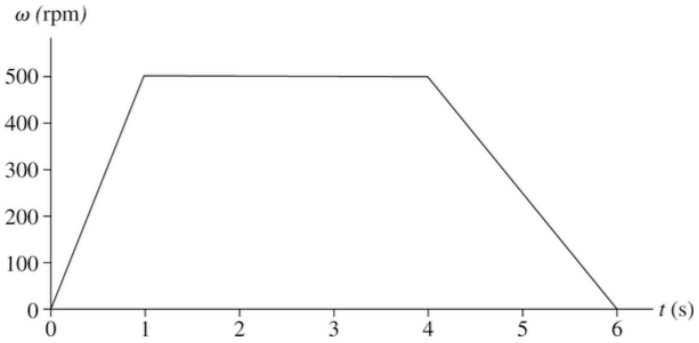
Converting, $77\ \text{rad} = 12.3$ revolutions.

REVIEW: Making a bicycle wheel turn just over 12 revolutions in 10 s when it is initially turning almost one revolution per second to begin with seems attainable by a cyclist.

3. Starting from rest, a DVD steadily accelerates to $500\ \text{rpm}$ in $1.0\ \text{s}$, rotates at this angular speed for $3.0\ \text{s}$, then steadily decelerates to a halt in $2.0\ \text{s}$. How many revolutions does it make?

MODEL: The DVD is a rotating rigid body.

VISUALIZE: The angular displacement (angle turned) is the area under the angular velocity graph.



SOLVE:

$$\Delta\theta = \tfrac{1}{2}(1\text{ s})(500\text{ rpm}) + (3\text{ s})(500\text{ rpm}) + \tfrac{1}{2}(2.0\text{ s})(500\text{ rpm}) = (4.5\text{ s})\left(500\ \frac{\text{rev}}{\text{min}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right) = 38\text{ rev.}$$

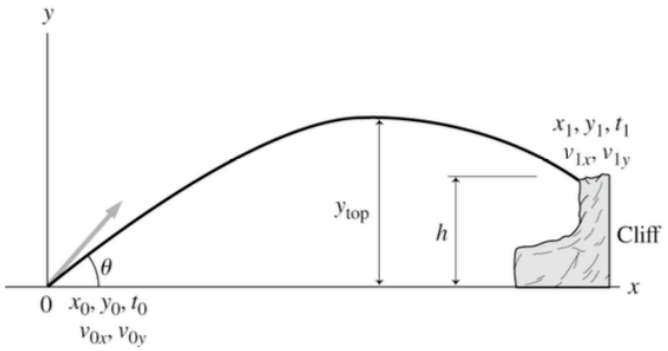
REVIEW: 38 revolutions seem like a reasonable amount in 6 s.

4. A ball is thrown toward a cliff of height h with a speed of $30\ \text{m/s}$ and an angle of 60° above horizontal. It lands on the edge of the cliff $4.0\ \text{s}$ later.
   a. How high is the cliff?
   b. What is the maximum height of the ball?
   c. What is the ball's impact speed?

MODEL: The particle model for the ball and the constant-acceleration equations of motion are assumed.

VISUALIZE:

Pictorial representation



Known
$x_0 = y_0 = t_0 = 0 \quad \theta = 60^\circ$
$v_0 = 30\ \text{m/s} \quad v_{1x} = v_{0x}$
$t_1 = 4.0\ \text{s}$
$a_y = -g$

Find
$y = h \quad y_{top}$
$v_{1x} \quad v_{1y} \quad v_1$

SOLVE: **(a)** Using $y_1 = y_0 + v_{0y}(t_1 - t_0) + \frac{1}{2}a_y(t_1 - t_0)^2$,

$$h = 0\ \text{m} + (30\ \text{m/s})\sin 60^\circ(4\ \text{s} - 0\ \text{s}) + \tfrac{1}{2}(-9.8\ \text{m/s}^2)(4\ \text{s} - 0\ \text{s})^2 = 25.5\ \text{m}$$

The height of the cliff is 26 m.

(b) Using $(v_y^2)_{top} = v_y^2 + 2a_y(y_{top} - y_0)$,

$$0\ \text{m}^2/\text{s}^2 = (v_0 \sin\theta)^2 + 2(-g)(y_{top}) \Rightarrow y_{top} = \frac{(v_0 \sin\theta)^2}{2g} = \frac{[(30\ \text{m/s})\sin 60^\circ]^2}{2(9.8\ \text{m/s}^2)} = 34.4\ \text{m}$$

The maximum height of the ball is 34 m.

(c) The x and y components are

$$v_{1y} = v_{0y} + a_y(t_1 - t_0) = v_0 \sin\theta - gt_1 = (30\ \text{m/s})\sin 60^\circ - (9.8\ \text{m/s}^2) \times (4.0\ \text{s}) = -13.22\ \text{m/s}$$

$$v_{1x} = v_{0y} = v_0 \cos 60^\circ = (30\ \text{m/s})\cos 60^\circ = 15.0\ \text{m/s}$$

$$\Rightarrow v_1 = \sqrt{v_{1x}^2 + v_{1y}^2} = 20.0\ \text{m/s}$$

The impact speed is 20 m/s.

REVIEW: Compared to a maximum height of 34.4 m, a height of 25.5 for the cliff is reasonable.